Integrated Methodology for Atmospheric Correction and Cloud Removal of Multispectral Remote Sensing Images Using Guided Diffusion Model

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***Abstract*— The atmospheric correction and cloud removal are core tasks of the atmospheric radiative transfer model in satellite remote sensing radiometric calibration. The accurate atmospheric correction and cloud removal of satellite remote sensing images are critical preliminary steps for effective satellite remote sensing applications. Current methods for atmospheric correction and cloud removal in satellite remote sensing imagery are mainly divided into traditional algorithms and deep learning methods. Traditional remote sensing algorithms typically involve complex atmospheric radiative transfer models, intricate meteorological parameters, synchronized ground measurements, and corresponding geographic data. There are significant difficulties and limitations in achieving detailed descriptions of ground and atmospheric models. Deep learning methods typically employ computer vision techniques. These methods cannot effectively utilize atmospheric parameters and multispectral image (MSI) features. Therefore, they cannot significantly improve the data quality of spectral satellite remote sensing images. This study integrates multitemporal and multispectral remote sensing images, atmospheric data (such as aerosol optical thickness and water vapor), and remote sensing data like surface reflectance (SR). A novel unified model is proposed to handle atmospheric correction and cloud removal for multispectral satellite remote sensing images. The unique geographical environment, and diverse and complex atmospheric conditions of the Guilin region make its satellite remote sensing data rich in research characteristics for atmospheric correction and cloud removal tasks. It can provide a typical and representative experimental scenario for this study. This study first utilized Sentinel-2 top-of-atmosphere (TOA) and SR data from the Guilin region to create four datasets: true color and multispectral fusion datasets with atmospheric correction incorporating aerosol parameters, single and multitemporal cloud removal datasets. Then, we developed a conditional guided diffusion model. This model can capable for both multispectral atmospheric correction and multitemporal cloud removal tasks by redesigning the conditional control and encoding sections. Our method achieved satisfactory results in both true color and multispectral atmospheric correction tasks. For atmospheric correction tasks on true color images (TCIs), our method achieved MAE, SSIM, and peak signal-to-noise ratio (PSNR) metrics of 0.0125, 0.9445, and 35.5221 dB, respectively. For the atmospheric correction task in MSIs, the method achieved MAE, SSIM, and PSNR metrics of 0.0174, 0.9499, and 36.3547 dB, respectively. For multitemporal true color cloud removal tasks, the method achieved MAE, SSIM, and PSNR metrics of 0.0224, 0.9556, and 36.2122 dB, respectively.**

***Index Terms*— Atmospheric aerosol, atmospheric correction, cloud removal, guided diffusion model, multitemporal and multispectral fusion, surface reflectance (SR).**

I. Introduction

CURRENTLY, the remote sensing imagery plays a crucial role in fields, such as geographic information analysis, urban planning, and environmental monitoring. The ultimate goal of satellite remote sensing radiometric calibration is to obtain high-precision and real-time remote sensing, which can more effectively support a variety of critical tasks. During the process of collecting satellite remote sensing imagery, the atmospheric interference from clouds and aerosols can significantly impact the quality and usability of surface image data. Therefore, the atmospheric correction and cloud removal are among the core tasks in the atmospheric radiative transfer model for satellite remote sensing radiometric calibration. It was found that atmospheric correction and cloud removal are closely related during research processes, and handling them separately often leads to suboptimal results.

This article designs a new integrated radiative transfer model. The method integrates deep learning with atmospheric correction models, specifically combining surface reflectance (SR) models under complex atmospheric and surface conditions to construct a novel remote sensing atmospheric

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radiative transfer model. This model can simultaneously perform atmospheric correction and cloud removal tasks to acquire accurate multispectral reflectance information of the Earth's surface (including both natural and artificial targets). The significance of this research is to address the impact of atmospheric conditions, clouds, and aerosols on the SR information received by satellites.

Atmospheric correction tasks for remote sensing imagery are typically divided into two categories: those affected by atmospheric aerosols, which can cause errors in the ground reflectance of the imagery. However, these data contain complete surface information [1], [2]. The other type is affected by cloud cover, resulting in the complete loss of surface information in the imagery. The aim of atmospheric correction and cloud removal is to restore the actual surface information from data impacted by cloud cover and atmospheric aerosols. This enhances the quality and usability of remote sensing imagery.

This study uses high-resolution Sentinel-2 multispectral remote sensing images from Guilin as the data source. It is a representative region in southern China. Two datasets were constructed for cloud removal and aerosol effect elimination, validating the effectiveness of the model.

The structure of this article is as follows. Section I provides the introduce of our method and related work. Section II provides a detailed introduction to the study area and the data sources selected for this study. Section III describes the dataset construction process, the principle of conditional guided diffusion model, and a detailed description of the model constructed in this study. Section IV introduces the model evaluation indicators and experimental results, providing detailed analysis and ablation experimental results; Section V provides the final summary of this experiment.

II. Related Works

In order to obtain high-quality, continuous, and timely remote sensing images, there has been significant progress in atmospheric correction, cloud detection, and removal algorithms in remote sensing image processing research. However, there are still challenges: accurately correcting atmospheric images, accurately correcting the influence of thin clouds, avoiding spectral distortion, and reconstructing high-quality information obscured by thick cloud cover [1], [2].

For data affected by atmospheric aerosols, common correction methods include image-based methods, methods based on radiative transfer models, or hybrid methods. Common methods involve applying dehazing algorithms from image processing to atmospheric correction, such as using dark channel prior (DCP) methods and their improved algorithms [3], [4], [5] or Haze optimized transformation (HOT) [6]. Methods based on radiative transfer models require first establishing models to estimate parameters of atmospheric aerosols, such as the 6S (Second Simulation of the Satellite Signal in the Solar Spectrum) model [7], MODTRAN [8], and other radiative transfer models, to perform atmospheric correction on remote sensing images based on various parameters. Hybrid methods [9], [10], [11] combine various dehazing and correction methods or add additional atmospheric measurement information to achieve better results in dehazing tasks. There has been extensive research on the processing of data affected by cloud cover, with numerous cloud removal methods proposed for remote sensing images. Common methods include using multispectral data, image enhancement, or using multitemporal data for cloud removal. For example, basic methods like pixel replacement [12], [13] use cloud-free regions from multiple images to replace cloud-covered areas. Other methods utilize image features, such as DCP algorithms/cite 14 or dark pixel methods [14], to remove thin clouds. In addition, research has shown that using radiative transfer models [11], [15], [16] can accurately handle thin clouds to better restore ground information.

Another common approach is to utilize multispectral data [17], which comes from sensors with different wavelengths, to restore information beneath cloud cover by exploiting differences in cloud-shadow images and spectral correlations. Some studies incorporate auxiliary information, such as using synthetic aperture radar (SAR) radar data [18], [19] with minimal cloud interference as ground auxiliary information for cloud removal.

In recent years, some new research directions have emerged in atmospheric correction and cloud removal tasks, including using deep learning methods for remote sensing cloud removal and dehazing tasks. Deep learning, as a powerful data-driven technique, has been widely applied in the field of image processing, achieving remarkable achievements. Traditional atmospheric correction and cloud removal algorithms are typically based on mathematical models and statistical methods, limited by complex optical phenomena and image characteristics, and their effectiveness may not be satisfactory. Deep learning methods can better understand features in images and provide more accurate processing of complex atmospheric and cloud cover situations through the learning and pattern recognition capabilities of deep neural networks. In remote sensing image processing tasks, deep learning offers new perspectives and methods for removing atmospheric interference and cloud cover. With the introduction of fully convolutional neural networks like U-Net [20], there has been a plethora of research based on convolutional neural networks in deep learning remote sensing image cloud removal and dehazing tasks [21], [22], [23], making full use of the image generation capabilities of fully convolutional neural networks to achieve cloud removal and dehazing tasks.

Some studies have used diffusion models for remote sensing cloud removal and dehazing tasks [24], [25], while others have proposed sequence-based diffusion models for cloud removal tasks in the remote sensing field, utilizing multimodal auxiliary information and temporal information for cloud removal tasks. Using a denoising diffusion probabilistic model (DDPM) architecture, cloud-covered optical images and auxiliary SAR images [26], [27] are used as input to provide important information for retrieving missing features.

For remote sensing cloud removal tasks, several publicly available datasets [27] have been constructed containing cloud and cloud-free samples of radar and optical observation

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data. In addition, datasets providing both thin and thick cloud removal have been created and published by Sarukkai et al. [23].

Although many studies on deep learning remote sensing image cloud removal and dehazing often use multispectral image (MSIs) or SAR data as auxiliary information to improve the effectiveness of cloud removal and dehazing of remote sensing images, they often overlook the influence of aerosols on the availability of processed images after cloud removal and dehazing. Aerosols significantly affect image quality by scattering, absorbing, or blocking light, resulting in image blur, decreased contrast, and color distortion. The presence of aerosols makes the interpretation and analysis of remote sensing images more challenging as they interfere with the transmission and reflection of light, affecting the recognition and measurement of surface features. Considering the influence of aerosols on remote sensing images is crucial for accurate identification and quantitative analysis of surface features. Aerosol optical depth (AOD) is a parameter describing the concentration and optical characteristics of aerosols in the atmosphere. It measures the absorption and scattering of aerosols on light and is an important indicator for evaluating the impact of aerosols on the quality of remote sensing images. Water Vapor pressure (WVP) is an indicator describing the water vapor content in the atmosphere, which is one of the important components of the atmosphere and significantly affects the transmission and reflection of light, especially in remote sensing image acquisition. However, there has been no effective use of aerosol data as auxiliary data in deep learning research. At the same time, remote sensing cloud removal and dehazing tasks are often segmented and implemented by different models, reducing the efficiency of image processing. Therefore, this study designs a conditional guided diffusion model to simultaneously perform cloud removal and atmospheric aerosol correction tasks. The model utilizes multispectral information, AOD, and WVP measurement data as model guiding conditions. Building on the enhancement of cloud removal and dehazing capabilities with multispectral data, this study further explores the impact of incorporating multispectral, aerosol, and water vapor data conditions into deep learning models on the effectiveness of cloud removal and dehazing of remote sensing images.

III. Materials

A. Research Area

The study area is situated in Guilin City, China, with geographical coordinates at approximately N25°16′35.35″ latitude and E110°17′7.52″ longitude. It encompasses a total area of approximately 27 700 km^2, with the core urban area covering about 866 km^2. The study area is depicted in Fig. 1. Guilin City features a diverse topography, mainly consisting of hilly terrain and small mountain basins. The urban areas are located on large plains within the Xianggui corridor. The southern part of the city has fragmented hilly plains, while expansive mountainous regions extend to the west, north, and east. The city is crossed by five rivers, including the Li River.

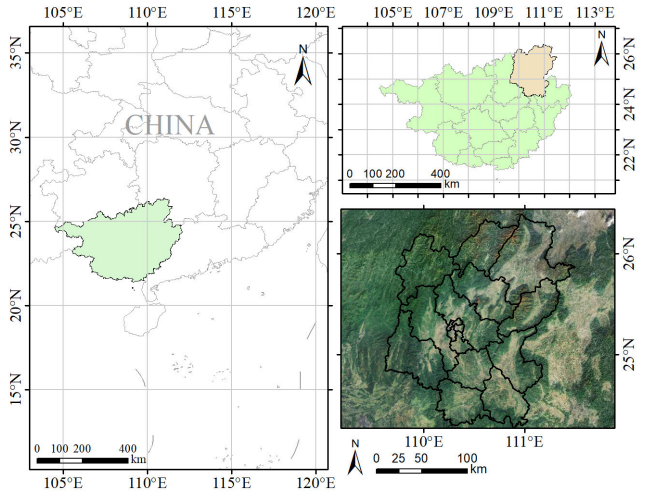


Fig. 1. Research area.

The average annual temperature in Guilin City is approximately 19.3 °C, with an annual rainfall reaching up to 1949.5 mm. The average evaporation fluctuates between 1490 and 1905 mm annually, while the average relative humidity remains between 73% and 79% throughout the year. The region is frequently affected by high humidity and substantial precipitation, leading to common occurrences of haze, extensive cloud cover, and atmospheric humidity issues, all of which adversely affect the quality of remote sensing images. The variability in meteorological conditions, including rainfall, fog, haze, and seasonal weather changes, significantly impacts the clarity and usability of remote sensing images. Therefore, the in-depth research into remote sensing image dehazing techniques specific to this region is crucial for effectively analyzing image data.

B. Data Source

This study utilizes Sentinel-2 multispectral satellite remote sensing data. The Sentinel-2 constellation consists of two satellites, Sentinel-2A and Sentinel-2B, orbiting at an altitude of 786 km in a sun-synchronous orbit. These satellites are equipped with the multispectral instrument (MSI), which captures high-resolution MSIs with a global revisit frequency of five days. The MSI onboard Sentinel-2 samples 13 bands covering the spectral range from 443 to 2190 nm. Specifically, there are four bands in the visible and near-infrared spectrum with a spatial resolution of 10 m, six bands in the red edge and shortwave infrared spectrum with a resolution of 20 m, and three bands in the atmospheric spectrum with a resolution of 60 m.

C. Data Collect

To research the potential impact of incorporating aerosol information on remote sensing image dehazing tasks. We integrated Sentinel-2 satellite's Level-1C top-of-atmosphere (TOA) reflectance products, which do not eliminate the influence of atmospheric aerosols, along with atmospherically corrected Level-2A SR products. We selected Level-2A SR as the baseline for atmospheric correction. In addition, to comprehensively consider aerosol and atmospheric conditions, aerosol

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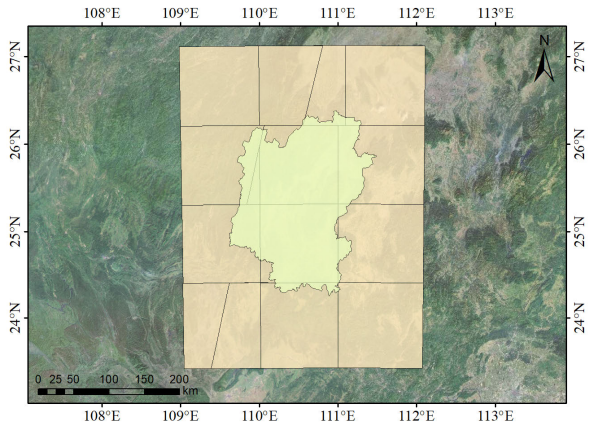


Fig. 2. Sentinel-2 tiles using in research.

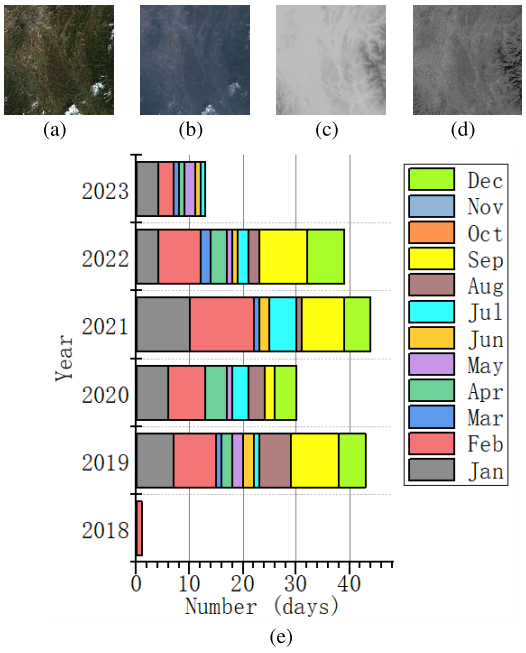


Fig. 3. Remote sensing data collection. (a) True-color imagery of Level-2A products. (b) TCIs of Level-1C products. (c) Visualization of AOD. (d) Visualization of water vapor. (e) Statistical distribution of image acquisition time.

optical thickness, and water vapor products were employed as auxiliary data. This integrated approach will contribute to a deeper understanding of the role of aerosol information in remote sensing image dehazing. This study covers the Guilin area with 14 tiles. The research scope of this dataset includes Guilin City and its surrounding areas, with a total area of approximately 155 000 km^2. The area is illustrated in Fig. 2.

In the selected research area mentioned earlier, this study chose Sentinel-2 satellite data Level-1C products with cloud coverage not exceeding 20% from January 2018 to September 2023. After filtering, a total of 170 days of image data were selected.

IV. Methods

We present a conditional guided diffusion model specifically for handling multispectral remote sensing images using Sentinel-2 satellite imagery. The model aims to effectively perform both MSI atmospheric correction and cloud removal tasks simultaneously. Unlike typical computer vision dehazing tasks, our study incorporates aerosol data as additional conditions for image generation in remote sensing image dehazing tasks.

The objective of the model in processing remote sensing images is to obtain the true ground reflectance information based on the remote sensing images and other atmospheric data. Furthermore, the model removes clouds from the remote sensing images and reconstructs the lost ground features obscured by clouds. The complete experimental process is illustrated in Fig. 4 and includes the following steps: 1) data acquisition; 2) dataset construction; 3) neural network model construction; and 4) model training and result evaluation.

A. Dataset

This study aimed to investigate the effectiveness of using aerosol information as auxiliary data in deep learning models for atmospheric correction and cloud removal. To achieve this aim, several datasets were constructed. The process of building datasets after obtaining Sentinel-2 remote sensing image products involves three main steps.

1) Preprocessing and mosaic.
2) Image clip and format.
3) Dataset formatting.

Fig. 5 illustrates the process of data processing and dataset construction. The final datasets comprise four sets for two tasks. Dataset 1 involves atmospheric correction of remote sensing images with aerosol data assistance and includes two sets of data: panchromatic images and multispectral SR. The objective is to train neural network models to output high-quality SR images. Dataset 2 involves cloud removal for multitemporal remote sensing images. The objective is to train neural network models to produce accurate representations of cloud-covered regions.

1) Dataset One: The atmospheric correction of remote sensing images with aerosol data assistance, including two sets of data: panchromatic images and multispectral ground reflectance, aiming to train neural network models to output high-availability ground reflectance images.

2) Dataset Two: Cloud removal from multitemporal remote sensing images, aiming to train neural network models to output high-fidelity representations of land areas obscured by clouds.

3) Preprocessing and Mosaic: The MSIs from the Sentinel-2 satellite have three different spatial resolutions: 60, 10, and 20 m. For this study, the images of bands B2–B8A, B11, and B12 from the Sentinel-2 L1C products were resampled to a spatial resolution of 60 m using the nearest neighbor method to ensure consistency in the dataset construction. In addition, we used the sen2cor tool to retrieve AOD and water vapor data as additional input conditions. This tool is based on the dense dark vegetation (DDV) algorithm and the atmospheric precorrected differential absorption algorithm (APDA, [3]). Our study utilized the L2A products as the standard values (ground truth) for atmospheric correction.

Then, we mosaic the downsampled images to form a complete multispectral remote sensing satellite image that covers the entire area of Guilin and its surrounding regions.

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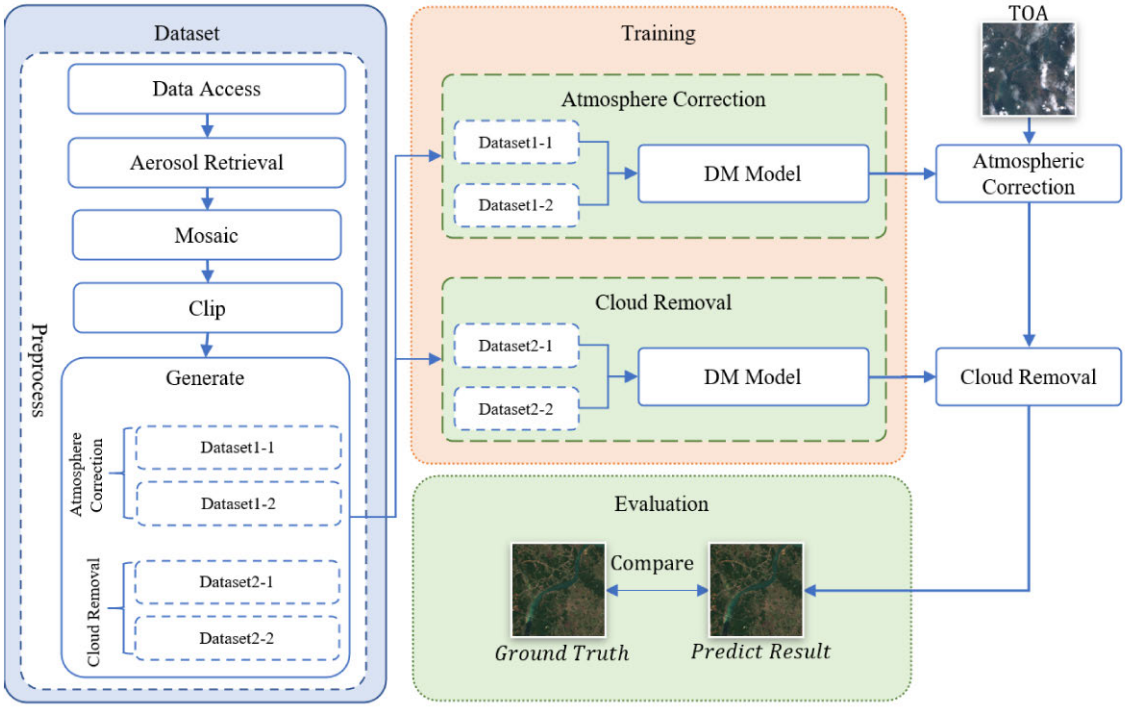


Fig. 4. Research overall flowchart.

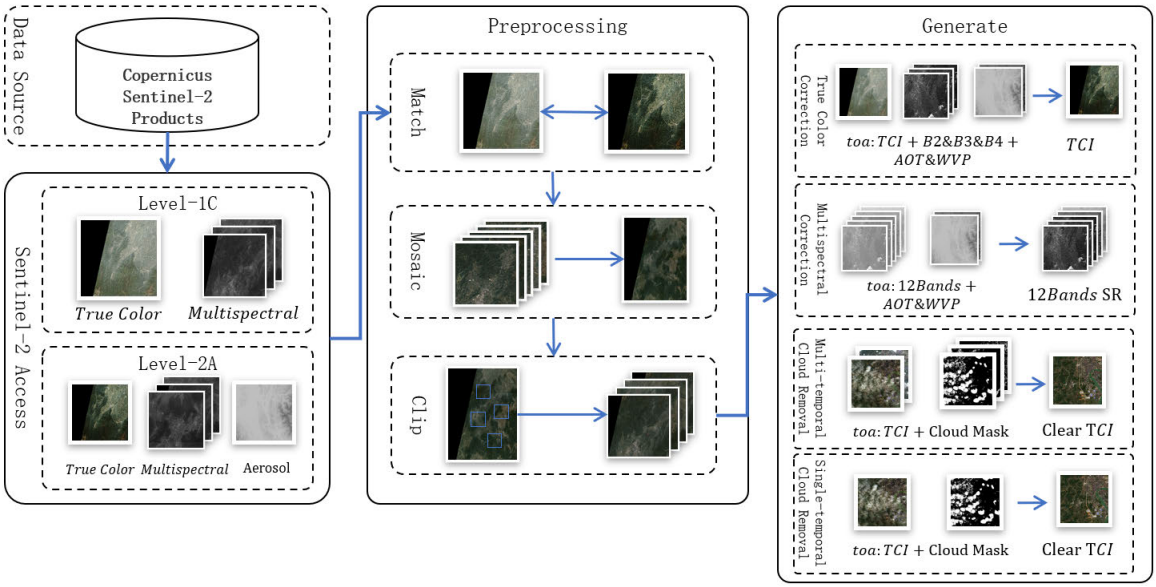


Fig. 5. Dataset preprocess and generate flowchart.

4) Image Clip and Format: To create the atmospheric correction dataset, we randomly crop areas from all 12 bands, TCI, and data on atmospheric aerosol optical thickness (AOT) and WVP. The resulting dataset consists of 5–30 images, each measuring 512 × 512 pixels. The random cropping operation excludes image boundary areas to ensure that the cropped regions contain complete reflectance, AOT, and WVP data.

The Sentinel-2 satellite has a revisit cycle of 5 days. To capture both cloudy and cloud-free images of the same location in the Guilin area, we also consider how changes in terrain might impact the cloud removal process. We use the cloud masks provided in the image products as selection criteria. We check for images acquired after the initial image within a one-month period. If both cloudy and cloud-free images can be obtained at the same location in this time series, they are paired together and added to the dataset.

In this study, we collected approximately 3000 images for atmospheric correction and about 500 sets of images for cloud removal. Each image is 512 × 512 pixels in size. Image samples are shown in Fig. 6. The atmospheric correction data includes TOA reflectance, SR, as well as AOT, and water vapor (WVP). The cloud removal data includes multitemporal panchromatic images and cloud masks. These datasets are split into training and validation sets in an 8:2 ratio for model training and evaluation. Specific dataset statistics are shown in Table I.

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Fig. 6. Example for a TCI.

TABLE I
DATASET STATISTICS

Datasets	Tasks	Size	Image Resolution
1-1 1-2	TCI AC MSI AC	3117	512×512
1-1 1-2	MT CR ST CR	586	512×512

Dataset 1-1: Dataset 1-1 is a TCI dataset. Each dataset includes three visible light bands: blue (B2), green (B3), and red (B4), with wavelengths of 496.6 nm (S2A)/492.1 nm (S2B), 560 nm (S2A)/559 nm (S2B), and 664.5 nm (S2A)/665 nm (S2B), respectively. The TCI is generated from TOA and SR reflectance. AOD and WVP data serving as auxiliary data for atmospheric correction. These data are used for model training. The TOA reflectance in the visible light bands, full-color images, AOT, and WVP data are utilized as input features for the model, while the surface full-color image serves as the learning target.

Dataset 1-2: Dataset 1-2 is a collection of 12 bands of MSIs, as well as AOD and WVP aerosol data. These data are utilized for model training, with the TOA reflectance MSI, AOD, and WVP data serving as input features. The corresponding band ground reflectance data are the learning targets for the model.

Dataset 2-1: Dataset 2-1 is a multitemporal dataset. It consists of several visible light band panchromatic images along with their corresponding cloud masks. Only the images with clouds are included in the multitemporal data. The data acquisition strategy is as follows.

1) Images with cloud mask coverage greater than 30% are considered as images with clouds.
2) Image from the same location with cloud coverage less than 2% are considered as cloud-free images and with cloud coverage greater than 15% as cloud-covered images.
3) Select the most recent cloudy image as input from several images.
4) Use the image that meets the criteria for cloud-free and has the most recent shooting time as the ground truth.

Dataset 2-2: Dataset 2-2 is a subset of Dataset 2-1 used exclusively to test single-time cloud removal performance. It includes only images with the highest cloud content as input and cloud-free images as the ground truth.

V. MODEL

The basic diffusion model denoising diffusion probabilistic models (DDPMs) have been widely used in various computer vision tasks, such as text-to-image generation and image restoration. Like other generative models, such as generative adversarial networks (GANs) and variational autoencoders (VAEs), diffusion models generate target data samples from random noise (using the Gaussian noise).

There are two processes in diffusion models: the forward process and the reverse process. The forward process is also referred to as the diffusion process. Both processes are based on parameterized Markov chains. It is used to represent the stochastic transition process from one state to another state within the state space.

The forward process adds noise to the sample image. This process gradually transforms the distribution of the original sample data x_0 into a standard Gaussian distribution x_T. Through multiple iterations, the noise gradually propagates and diffuses throughout the entire sample. The reverse process restores the data from a Gaussian distribution back to the distribution of the original samples by denoising steps. The goal is to retrace the path of noise addition for data generation. Samples generated by this process help to evaluate the performance of the denoiser and perform other tasks.

A. Diffuse

The diffusion process gradually adds Gaussian noise to image data until the input image content becomes random noise. In DDPM, this forward process involves cumulatively adding Gaussian noise to each step's input data x_i over T steps. It can be expressed as follows:

$$q(X_{1:T}|X_0) := \prod_{t=1}^{T} p\theta(X_t \mid X_{t-1})$$
$$q(X_t \mid X_{t-1}) := \mathcal{N}\left(X_t; \sqrt{1-\beta_t}X_{t-1}, \beta_t I\right). \quad (1)$$

The expression $q(X_{1:T} \mid X_0)$ represents the conditional probability distribution of the variable sequence $X_{1:T}$ from time step 1–T, given the initial state X_0. $\prod_{t=1}^{T} p_\theta(X_t \mid X_{t-1})$ denotes the product of conditional probabilities from time step 1–T.

$q(X_t \mid X_(t-1))$ gives the conditional probability of X_t at time t given $X_(t-1)$ at step $t-1$. $N(X_t; (1-\beta_t)^{1/2}X_(t-1), \beta_t I)$ is a Gaussian with mean $X_(t-1)$ and variance β_t. Each step of the diffusion process generates X_t, and the process can be represented as a sequence of these conditional distributions

$$q(x_{1:T} \mid x_0) = \prod_{t=1}^{T} q(x_t \mid x_{t-1}). \quad (2)$$

B. Denoise

The reverse diffusion model uses a neural network to estimate noise. It uses network to restore data sampled from a standard Gaussian to the original sample distribution by predicting the noised distribution. The reverse process is

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formulated as follows:

$$p(X_{0:T}) := p(X_T)\prod_{t=1}^{T} p\theta(X_{t-1} \mid X_t) \tag{3}$$

$$p(X_{t-1} \mid X_t) := \mathcal{N}\left(X_{t-1}; \mu_\Theta(X_t, t), \sum_\Theta(X_t, t)\right). \tag{4}$$

$q(X_{0:T})$ represents the probability distribution of the variable sequence $X_{0:T}$ from time step 0 to T. $p(X_T)$ is the probability distribution of X_T at time step T. $\prod_{t=1}^{T} p_\theta(X_{t-1} \mid X_t)$ denotes the product of conditional probabilities from time step 1 to T. Each $p_\theta(X_{t-1} \mid X_t)$ represents the probability distribution of X_{t-1} at time step $t-1$, given X_t at time step t.

$N(X_{t-1}; \mu_\Theta(X_t, t), \Sigma_\Theta(X_t, t))$ represents the probability density function of the multivariate Gaussian distribution generated by the input functions μ_Θ and $\Sigma_\Theta(X_t, t)$, where $\mu_\Theta(X_t, t)$ denotes the mean vector and $\Sigma_\Theta(X_t, t)$ denotes the covariance matrix. This Gaussian distribution's probability density function describes the possible values of X_{t-1} given the input X_t and time step t.

By training the neural network, we can learn the mean and variance for each time step of the inverse diffusion process. These learned parameters dynamically generate Gaussian distributions based on the input X_t and time step t, used for inferring the original distribution of noisy data.

C. Condition

The diffusion model is designed to acquire image generation capabilities based on denoising and restoration processes in image tasks. However, conventional diffusion models almost entirely lose the original information of the input image by diffusing it into Gaussian noise, resulting in random model outputs. This does not meet the requirements for image atmospheric correction. Hence, we employ the classifier-free guidance diffusion model (CFGDM) [28]. It utilize a conditional guidance approach. Conditional guidance is designed for aerosol and spectral data to guide the model during neural network model training.

The reverse process of the diffusion model $p(X_{t-1} \mid X_t)$ can infer the original image from Gaussian noise images. The reverse process becomes inferring X_{t-1} under the conditions of X_t and y when y is taken as a model condition. The formula is as follows:

$$p(X_{t-1} \mid X_t, y). \tag{5}$$

The variance is treated as a constant in DDPM and the neural network is solely used to fit the mean $\mu_\theta(X_t, t)$. Therefore, the neural network model designed in this study will modify the reverse process as follows:

$$p(X_{t-1} \mid X_t) := \mathcal{N}\left(X_{t-1}; \mu_\Theta(X_t, t, y), \sum_\Theta(X_t, t)\right). \tag{6}$$

The denoising model typically constructs the reverse denoising process of the diffusion model using a neural network architecture similar to U-Net. It is used to predict the noise distribution based on guiding conditions, such as noisy images and time steps. U-Net is a commonly used convolutional neural network structure primarily employed in tasks, such as image semantic segmentation and image generation. It features an "encoder–decoder" architecture, which extracts features and retains large-scale information of images through the processes of downsampling and upsampling.

The structure of U-Net consists of two parts: the left part and the right part. The left part represents the encoding process, responsible for downsampling and feature extraction, reducing the spatial dimensions of feature maps while preserving important features. This allows for the gradual reduction of image size while extracting features. The right part represents the decoding process, primarily used for upsampling and feature fusion. Each level performs upsampling operations. The obtained feature map is concatenated with the corresponding feature map on the left side to fuse multiscale feature information after each upsampling operation. This gradually restores the image size and increases the number of channels. Finally, the U-Net network typically includes an output layer. It utilizes a convolutional network module to adjust the output structure to match the format of the target output.

D. Strategy and Loss

DDPM gradually transforms images into random noise during the diffusion process, and the representation of image features gradually approaches a Gaussian distribution. DDPM employs Markov chains to model the evolution process of images. A Markov chain is a stochastic process that describes the transition from one state to another in an image. Neural networks are used to represent the probability distribution of image features. Typically, the U-Net structure is used for feature extraction and reconstruction of image data. In addition, the conditional generation (denoising) process depends on the neural network training, with a given set of conditions, such as time steps and reference images. The diffusion model obtains suitable mean $\mu_0(X_t, t)$ and variance $\sum_\Theta(X_t, t)$ through the neural network training. The model's maximum likelihood estimation is obtained through variational inference, with its variational bound L computed using the KL divergence. The optimization method based on the variational bound can be expressed as follows:

$$E[-\log p_\theta(x_0)] \leq E_q\left[-\log \equiv \frac{p_\theta(x_{0.T})}{q(x_{1.T} \mid x_0)}\right]. \tag{7}$$

In the reverse diffusion process of DDPM, the variance is set to a fixed value, and the model only needs to predict the mean. The KL divergence of the normal distribution can be transformed into the mean squared error (MSE). DDPM adjusts the network's prediction of noise rather than the distribution mean, using the MSE by comparing the posterior Gaussian noise and the model-predicted noise as the loss

$$\text{MSE} = \frac{\sum_{i=1}^{m}(y_i - \hat{y}_i)^2}{m}. \tag{8}$$

E. Denoise Network

The objective of our research is to utilize deep learning methods to implement an atmospheric correction model. The

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performance of this model will be validated in both atmospheric correction and cloud removal tasks. In addition, the effects of multispectral remote sensing data and atmospheric aerosol data as additional conditions on atmospheric correction and cloud removal will be further investigated.

This study is based on the encoder–decoder architecture of U-Net. We utilize the multispectral remote sensing images and atmospheric data as guidance conditions and design a new denoising model. The model incorporates newly designed encoder and decoder modules on the top of the U-Net model. In addition, the model includes an additional conditional guidance pathway. The overall structure of the model is illustrated in Fig. 7.

1) Condition Guide Module: In order to accomplish the task of remote sensing image defogging under conditional guidance, this study incorporates a conditional encoding module into the model. This module receives multispectral remote sensing images and aerosol parameters as external guiding conditions. The conditional encoding module utilizes a three-layer deep skip convolutional module to learn features. These conditional features are then concatenated with the main backbone network of the model through a downsampling module, as depicted on the left side of Fig. 7(a).

2) Downsample Module: A convolutional module is employed to process the upper level input in the input section of the encoding network. A deep network is utilized to handle time embedding. The downsampling module incorporates residual and attention mechanism modules. Subsequently, the output features are passed through the downsampling module.

3) Upsample Module: The upsampling module employs the features decode module to fuse the upsampled results from the previous step with the features connected via skip connections from the encoding section and utilizes the Conv2DTranspose module for upsampling. The module is illustrated in Fig. 7(a).

F. Evaluation Metrics

To quantitatively analyze the performance of the model in atmospheric correction and cloud removal tasks for remote sensing images. This study employed six evaluation metrics: mean absolute error (MAE), structural similarity index (SSIM), peak signal-to-noise ratio (PSNR), root mean square error (RMSE), Pearson correlation coefficient, and spectral angle mapper (SAM) to objectively assess the similarity between images generated by the model and ground truth.

1) Mean Absolute Error: Use to evaluate the gap between the model's predicted values and the ground truth, assessing the quality of the model's output images. The smaller the difference from the ground truth, the higher the usability of the model's output images

$$\mathrm{MAE}(x, y) = \frac{1}{n}\sum_{i=1}^{n}|y_i - f(x_i)|. \tag{9}$$

2) Root Mean Square Error: Used to evaluate the difference between the model's predicted values and the ground truth, and is more sensitive to outliers. The formula is as follows:

$$\mathrm{RMSE}(x, y) = \sqrt{\frac{1}{n}\sum_{i=1}^{n}\big(y_i - f(x_i)\big)^2}. \tag{10}$$

3) Peak Signal-to-Noise Ratio: PSNR evaluates the difference between the model's predicted values and the ground truth. It evaluates the image quality with a focus on contrast comparison, measured in decibels (dB). A higher value indicates less difference in image quality. In the formula, MAX_I represents the maximum pixel value calculated according to the image bit depth

$$\mathrm{PSNR} = 10 \cdot \log_{10}\left(\frac{\mathrm{MAX}_I^2}{\mathrm{MSE}}\right). \tag{11}$$

4) Structure Similarity Index Measure: The SSIM result is expressed as a value between 0 and 1, with higher values indicating a smaller difference between the output image and the undistorted image. The model primarily measures based on luminance, contrast, and structure. In the formula, x and y represent the input comparison images, and α, β, and γ are the scaling factors, all set to 1 here. μ represents the mean and σ represents the standard deviation

$$\begin{aligned}\mathrm{SSIM}(x, y) &= [l(x, y)]^{\alpha}[c(x, y)]^{\beta}[s(x, y)]^{\gamma}\\ l(x, y) &= \frac{2\mu_{x\mu_y} + c_1}{\mu_x^2 + \mu_y^2 + c_1}\\ c(x, y) &= \frac{2\sigma_{xy} + c_2}{\sigma_x^2 + \sigma_y^2 + c_2}\\ s(x, y) &= \frac{\sigma_{xy} + c_3}{\sigma_x\sigma_y + c_3}.\end{aligned} \tag{12}$$

5) Pearson Correlation: The Pearson correlation is used to evaluate the correlation between model predicted values and actual values, where cov represents the covariance and σ represents the standard deviation

$$R = \frac{\mathrm{cov}(X, Y)}{\sigma_X\sigma_Y}. \tag{13}$$

6) Spectral Angle Mapper: Spectral angle measures the similarity between spectral features by computing the angle between them. It ranges from -1 to 1, where 1 indicates complete similarity and 0 indicates independence

$$\mathrm{SAM} = \arccos\left(\frac{X \cdot Y}{|X| \cdot |Y|}\right). \tag{14}$$

VI. Result

In this study, we utilized deep learning to develop atmospheric correction and cloud removal models. Our research focused on Guilin City and its surrounding areas in Guangxi Province, China. To validate the performance of the same model in both atmospheric correction and remote sensing image cloud removal tasks, we curated several datasets. Furthermore, we conducted an additional investigation into the influence of introducing atmospheric aerosol observation data as an additional condition on the effectiveness of the model's image processing.

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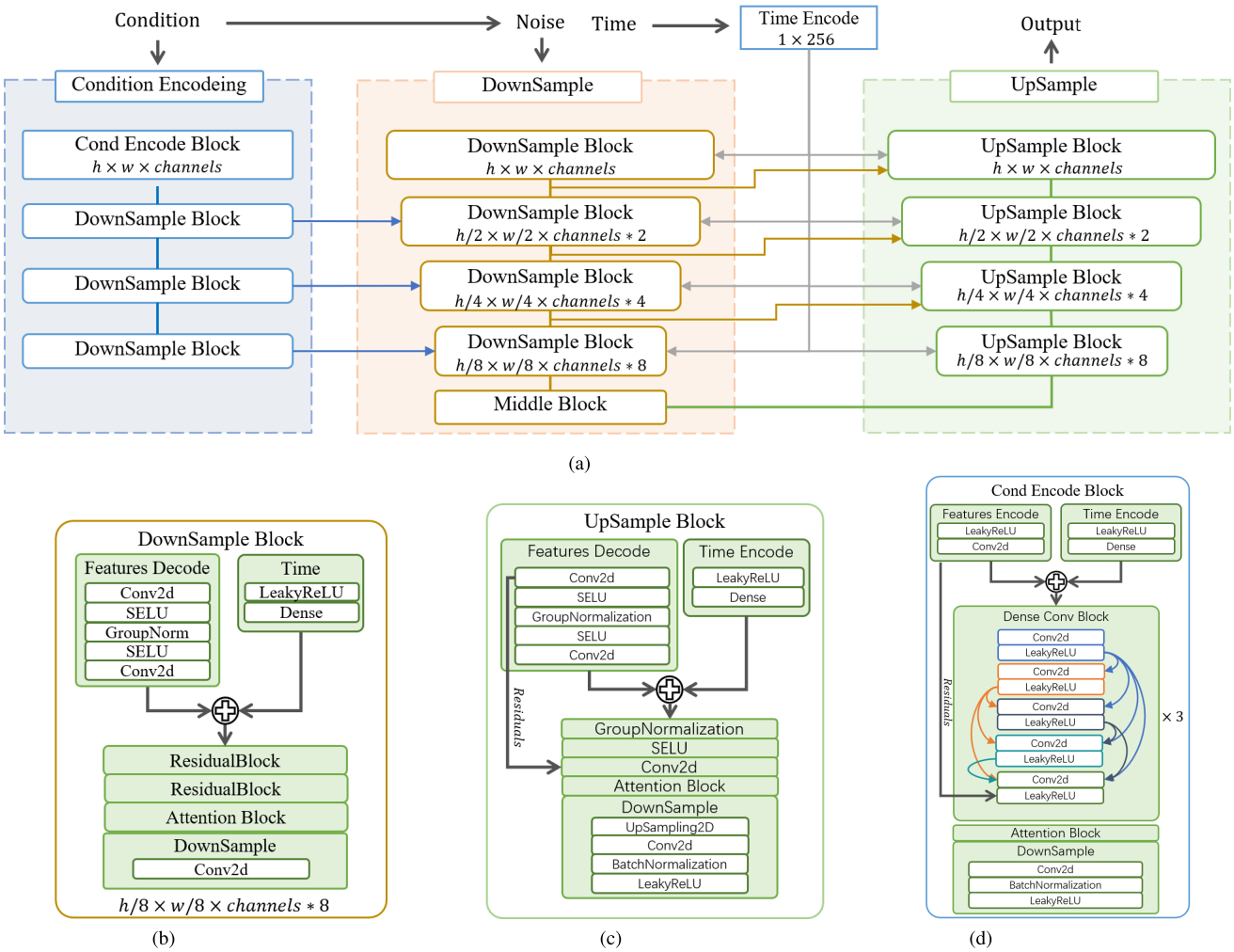


Fig. 7. Structure of the neural network. (a) Network. (b) Downsample. (c) Upsample. (d) Condition encode.

A. Parameters

We used the satellite remote sensing dataset created in our study for the Guilin region. In terms of dataset preprocessing, the image size of the atmospheric correction dataset was set to 512 × 512, and the data were normalized to [−1.0, 1.0]. Cloud removal dataset images were sized at 256 × 256 and normalized to [−1.0, 1.0]. The experimental data were divided into training and validation sets at a ratio of 70% and 30%, respectively. After training the deep learning models, performance metrics were evaluated on the validation set. Nondeep learning methods directly processed images on the validation set for performance evaluation.

Our study trained three models separately: haze removal for panchromatic images, atmospheric correction for MSIs, and declouding for multitemporal remote sensing images. All models utilized the same neural network architecture, with only the input conditions and output scales modified.

The experimental parameters for model training were consistent across the four datasets constructed and for tasks involving both panchromatic and MSIs. Model settings included a diffusion step size of 1000 steps, utilizing the DDIM optimization model with a denoising step size of 50 for all tasks. The training process employed the AdamW optimizer. We apply a warmup strategy with an initial learning rate of 1e−9 and linearly increasing to 1e−4 over 10% of the total training epochs. The learning rate was set to 1e−4 with EarlyStop enabled when training started. Experimental results were obtained either upon triggering EarlyStop or upon completion of training.

B. Atmosphere Correction Result

The atmospheric correction was trained using datasets 1-1 and 1-2, and the model's performance metrics were tested on the validation set. Fig. 8 illustrates the processing effects of the model on several typical land surface panchromatic images within the study area. From top to bottom, the figure displays TOA, ground truth (panchromatic images synthesized from multispectral SR images), and the model's atmospheric correction results.

Three types of panchromatic images are presented as experimental results in Fig. 8. It includes images with vegetation as the main cover, urban areas as the primary regions, and areas including water bodies.

For an accurate comparison of model performance. We conducted a quantitative evaluation using three evaluation metrics: MAE, SSIM, and PSNR, on the test set. The overall

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Fig. 8. Example of atmospheric correction results.

TABLE II
QUANTITATIVE EVALUATION RESULTS OF ATMOSPHERIC CORRECTION

Area	Quantitative Metrics					
	TCI Correction			MSI Correction		
	MAE	SSIM	PSNR	MAE	SSIM	PSNR
Vegetation	0.0341	0.7162	26.7128	0.0323	0.5968	27.0577
Urban	0.0125	0.9445	35.5221	0.02395	0.9624	31.7181
Water	0.015	0.9667	33.9455	0.0089	0.9756	38.9972
Overall	**0.0125**	**0.9445**	**35.5221**	**0.0174**	**0.9499**	**36.3547**

accuracy levels of our method on the true color image (TCI) correction and MSI test sets are presented in Table II.

Our proposed model achieved an average SSIM of 0.94 and a PSNR of 35.5 dB when comparing the corrected images with ground truth in the panchromatic image (TCI) correction task. In the MSI correction task, our model obtained an average SSIM of 0.94 and a PSNR of 36.3 dB. To further investigate the model's correction capability for different land surface features, we proceeded to individually test mountainous areas, urban areas, and water bodies within cloud-free regions. The specific results are detailed in Table II.

We first selected the mountainous areas in the northern part of Guilin to analyze the correction performance in these regions, as depicted in Fig. 9(a). Fig. 9(b) illustrates the correction discrepancies in mountainous areas, and the MAEs of the correction results from different algorithms are visualized.

Mountainous remote sensing images typically encompass various regions, including dense vegetation cover, bare ground, and water bodies, exhibiting complex textures and reflectance characteristics. Upon subjective evaluation of correction results in mountainous areas, our model demonstrates good performance. Compared with ground truth, the model effectively restores textures in mountainous, water, and bare ground areas. However, there is notable atmospheric correction error in the vegetation-covered areas north of the ridge line. As shown in Fig. 12(b)'s difference heatmap, continuous bright patches are observed in the ridge area. The reflectance error reaches 0.3, covering approximately 5% of the sampled area.

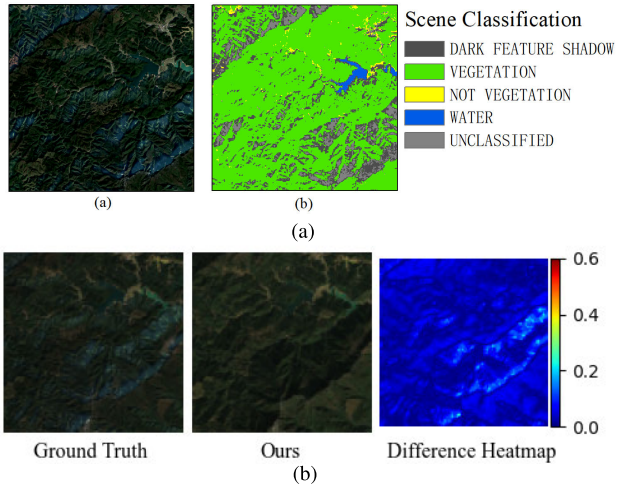


Fig. 9. Atmospheric correction results for vegetation area. (a) Selected region and land cover classification. (b) Mean errors.

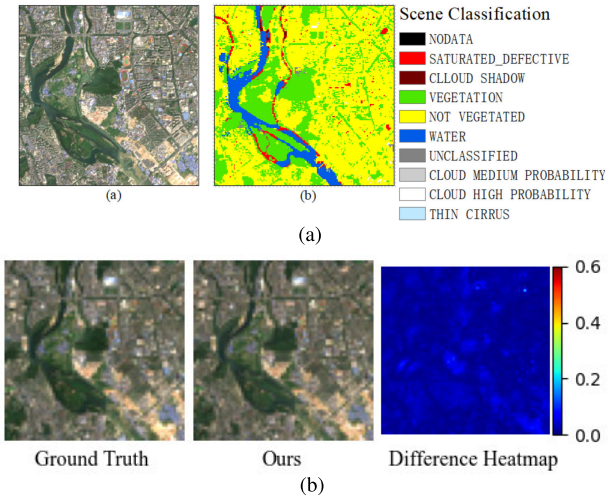


Fig. 10. Atmospheric correction results for urban area. (a) Selected region and land cover classification. (b) Mean errors.

Urban areas typically comprise diverse land cover types as illustrated in Fig. 10(a). It often includes buildings, plazas, road networks, water bodies, green spaces, and various other features within relatively small areas. Consequently, atmospheric correction tasks in urban regions demand a

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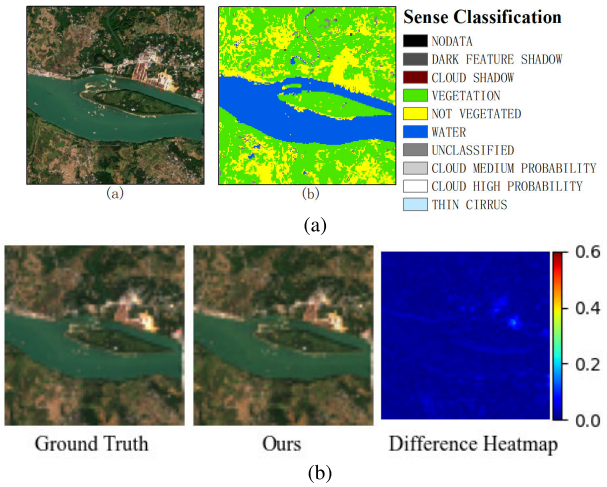


Fig. 11. Atmospheric correction results for water area. (a) Selected region and land cover classification. (b) Mean errors.

higher level of precision in restoring ground reflectance characteristics by the model. We conducted tests on remote sensing images of urban areas in Guilin. The urban area is illustrated in Fig. 10(b) as an example. The results of atmospheric correction on the panchromatic image (combined bands B2–B4) are shown in Fig. 15, where our method achieves the closest approximation to the ground-truth reflectance image.

Our method achieved an SSIM of 0.98 and a PSNR of 33.9. It demonstrates particularly good performance in restoring ground textures, especially in areas, such as buildings and road networks. In addition, the atmospheric correction effect is closest to the ground-truth reflectance image, with an absolute mean error of approximately 1.8.

The atmospheric correction results for water body areas are exemplary, as demonstrated by the imagery featuring rivers and riverbanks in Fig. 11(a). Our method achieved optimal results, which are presented in Fig. 17. Our method closely matches the reflectance correction of river water surfaces to the actual ground-truth imagery and effectively corrects the surrounding riverbank areas, including buildings and vegetation. Compared with the ground-truth reflectance, our method maintains an overall MAE of 1.5%. Our method achieves an SSIM of 0.96 and a PSNR of 33.9.

Our method was utilized to correct the 12 bands of Sentinel-2 imagery in our task of atmospheric correction for multispectral remote sensing images. This includes the B1 (442 nm) aerosol band, the visible light bands from B2 (490 nm) to B4 (665 nm), the red edge bands from B5 (705 nm) to B7 (783 nm), the B8 (842 nm) near-infrared band, and the shortwave infrared bands from B10 (1375 nm) to B12 (2190 nm). Overall, our method achieved an SSIM of 0.94 and a peak signal-to-noise ratio (PSNR) of 36.3547 dB.

To effectively showcase the performance of our method. We chose urban, water, and vegetated areas in the southern part of Guilin City and along the Li River basin for detailed comparison with ground truth reflectance data. The areas selected are illustrated in Fig. 12.

1) *Urban Area:* Covering the eastern urban area of Guilin City. This region is used for comparing the multispectral atmospheric correction results in urban regions. The statistical analysis of multispectral SR in this area is shown in Fig. 12(b).
2) *Mountainous and Vegetated Area:* Covering the mountainous areas along the Li River bank. It is used for comparing the atmospheric correction effects in mountainous and vegetated regions. The statistical analysis of multispectral ground reflectance in this area is shown in Fig. 12(c).
3) *Water Body Area:* Including the Li River and its adjacent areas. It is used for comparing the atmospheric correction results for water bodies. We extracted the river area after land cover classification (SC) and conducted a statistical analysis of ground reflectance, as shown in Fig. 12(d).

TABLE III
QUANTITATIVE EVALUATION RESULTS OF ATMOSPHERIC CORRECTION FOR EACH BAND

Band	MAE	RMSE	SSIM	PSNR	R	SAM
B01	0.0236	0.0251	0.8961	32.018	0.9415	0.1522
B02	0.0264	0.0277	0.8843	31.1581	0.9627	0.1634
B03	0.0214	0.023	0.9389	32.7689	0.9596	0.1026
B04	0.0212	0.0226	0.9262	32.9302	0.9824	0.138
B05	0.0218	0.0238	0.9509	32.4584	0.9631	0.0771
B06	0.0177	0.0232	0.9437	32.6778	0.9414	0.0624
B07	0.0186	0.0253	0.9445	31.9239	0.9486	0.0617
B08	0.0666	0.0753	0.872	22.4633	0.8515	0.1092
B09	0.0171	0.0222	0.9408	33.0888	0.9577	0.0521
B11	0.0188	0.0225	0.955	32.9539	0.966	0.0571
B12	0.0217	0.0236	0.9633	32.5499	0.9859	0.0729
B8A	0.0192	0.0261	0.9444	31.6508	0.9536	0.0595
Overall	**0.0245**	**0.0284**	**0.93**	**31.5535**	**0.9512**	**0.0924**

The quantitative evaluation of atmospheric correction within this area is presented in Table III. Fig. 14 illustrates the differences between the original and corrected data. Our method achieved an average MAE of 0.025, RMSE of 0.029, and a correlation coefficient of approximately 95% in the processing of multispectral bands within this region. Fig. 3 illustrates the correlation between the atmospheric correction results and standard values for three sampled areas. It can be observed that the majority of points from our method align closely with the 1:1 line, particularly in urban and water body areas.

To further analyze the performance of our method, we conducted separate tests on the atmospheric correction effects in urban, mountainous, vegetated, and water body regions. These regions are depicted in Fig. 12(a). We measured the average ground reflectance values across 12 bands in these areas and compared them with the ground-truth reflectance. To provide more detailed atmospheric correction results, we selected 50 sample points for further comparison. The following bands were selected as examples for analysis in plots: B2–B4, B6, B8A, and B12.

Fig. 14(a) presents the multispectral atmospheric correction results in the urban area of Guilin City. The spectral reflectance curve shown in Fig. 14(a) closely resembles the standard curve of ground reflectance. Our method achieved an overall MAE of 0.026, a correlation coefficient of 98%, and a SAM of 0.066. The reflectance statistics of sampled points in Fig. 15

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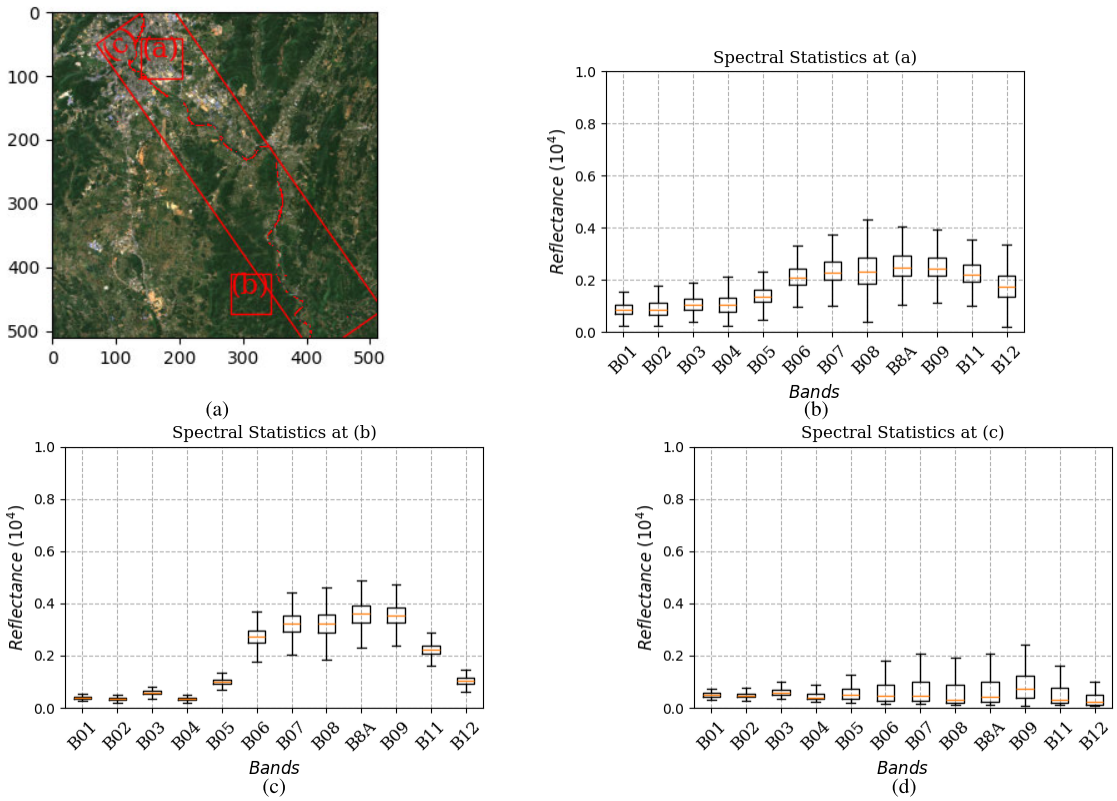


Fig. 12. Selected area spectral characteristics. (a) Area selection. (b) Spectral characteristics of urban areas. (c) Spectral characteristics of vegetation-covered areas. (d) Spectral characteristics of water bodies (Li River).

indicate that our method closely adheres to the standard values in the near-infrared and red-edge bands. There are slight overestimations in the shortwave infrared and RGB bands, but the average error is less than 2.7%, with an average SAM of approximately 0.05.

The atmospheric correction results for the vegetated areas along the Li River basin in Guilin City are illustrated in Fig. 14(b). Our method yields results that closely align with the standard reflectance curve, achieving an overall MAE of 0.028, a correlation coefficient of 91%, and a SAM of 0.085. However, there are some errors in the B7 and B8 bands, with an average error of approximately 9% and a SAM of 0.08.

The atmospheric correction results for the water body areas along the Li River in the southern part of Guilin City are shown in Fig. 14(c). Our method closely matches the standard reflectance curve. Overall, the MAE is 0.027, with a correlation coefficient of 93% and a SAM of 0.087. In general, compared to the standard values, our method shows a slight overestimation of approximately 2.3% in the B01 and visible light bands and a slight underestimation of about 7% in B08.

C. Cloud Removal

We use the multitemporal remote sensing cloud removal dataset (dataset 2-1) in Guilin City to test our method. The collection area covers urban, mountainous and river areas. Test images include thin clouds (which do not completely obscure the ground) and full-color image slices with cloud coverage between 15% and 100%. Also, we used the image sequence with length = 4 for testing, and the results are shown in Fig. 18.

TABLE IV
QUANTITATIVE EVALUATION RESULTS FOR CLOUD REMOVAL

Method	Quantitative Metrics		
	MAE	SSIM	PSNR
Ours	0.0224	0.9556	36.2122

Fig. 18 illustrates the cloud removal results obtained using a time series of length 4. Due to the stable revisit cycle of satellite remote sensing images over the same area, some images in the sequence contain almost no cloud cover (coverage less than 2%). It allows the model to accurately infer cloud-obscured areas and replace them. This typically yields optimal results for such multitemporal data. Our model can still effectively infer the ground features in the obscured areas when the sequence of images is entirely covered by thick or thin clouds. The results are shown in Fig. 18(d) or (f). Our method achieves approximately 0.97 SSIM and 32 dB PSNR in the multitemporal cloud removal task when use length = 3. The best results are achieved when Length = 4, with approximately 0.95 SSIM and 36 PSNR, as shown in Table IV.

D. Atmospheric Corrections and Cloud Removal Composite Results

In this section, we use the models trained in Sections VI-B and VI-C for comprehensive remote sensing image processing. The atmospheric correction model uses several sets of TOA images from the corresponding positions in Dataset 2-1 as

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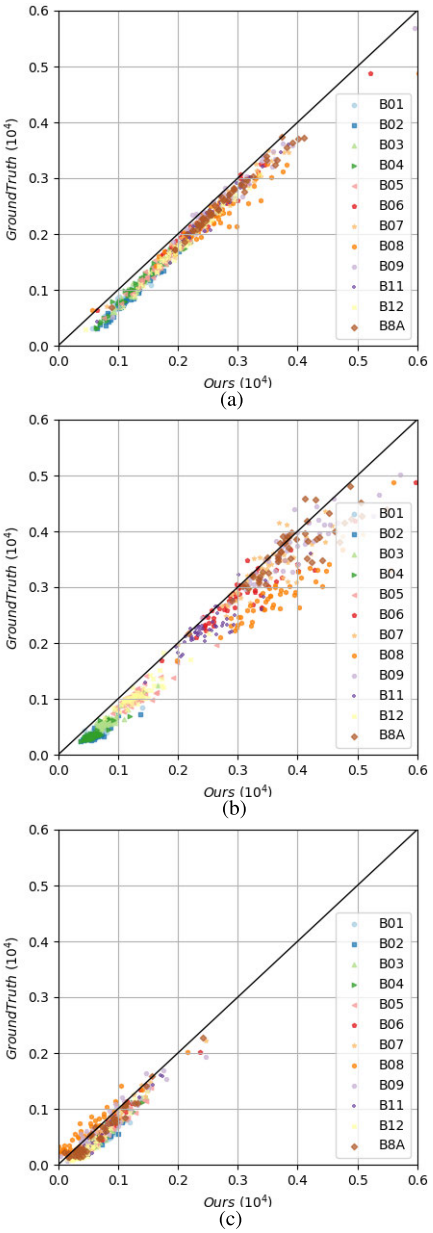


Fig. 13. 1:1 line, comparing the atmospheric correction results of our method with the ground truth. (a) Urban area result. (b) Vegetation area result. (c) Water area result.

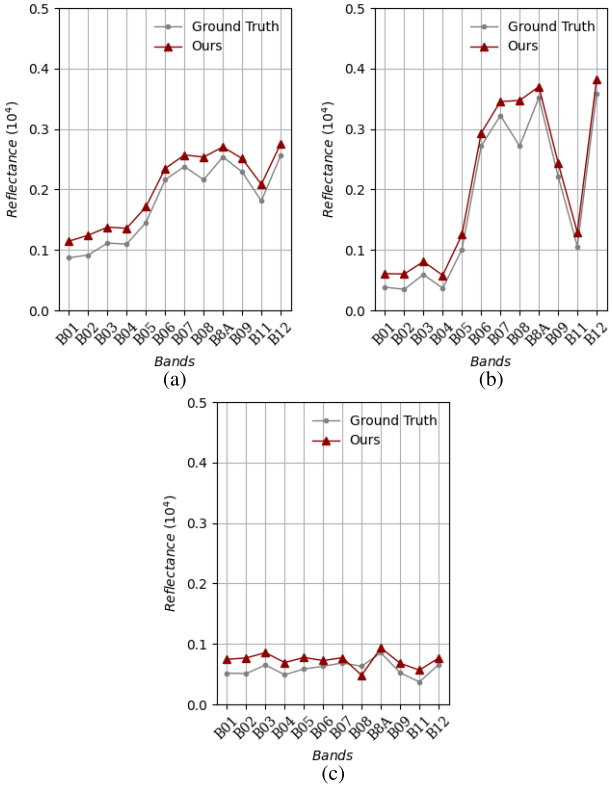


Fig. 14. Comparison of multispectral mean reflectance with ground truth. (a) Area A. (b) Area B. (c) Area C.

TABLE V

QUANTITATIVE EVALUATION RESULTS OF ATMOSPHERIC CORRECTION AND CLOUD REMOVAL WITH MAE, SSIM, AND PSNR

Step	Quantitative Metrics		
	MAE	SSIM	PSNR
Atmospheric Correction	0.0151	0.9335	34.0215
Cloud Removal	0.0155	0.9495	33.5791

TABLE VI

QUANTITATIVE EVALUATION RESULTS FOR AEROSOL PARAMETER ABLATION EXPERIMENT

Target	Strategy	Quantitative Metrics		
		MAE	SSIM	PSNR
True Color	Without AOD&WVP	0.0188	0.9303	32.7515
	With AOD&WVP	0.0125↑	0.9445↑	35.5221↑
Multi Spatial	Without AOD&WVP	0.0584	0.8329	26.8269
	With AOD&WVP	0.0174↑	0.9499↑	36.3547↑

inputs. The cloud removal model takes atmospheric correction results with length = 4 as inputs to complete the cloud removal. The specific process is illustrated in Fig. 19.

The experimental results are shown in the table. This study effectively handles atmospheric correction for multiple images by combining atmospheric correction and cloud removal models. The SR images after atmospheric correction are used for multitemporal cloud removal, resulting in cloud-free SR images for the target area. Compared with the ground truth in Dataset2-1. The final cloud removal results achieved MAE of 0.0155, SSIM of 0.9495 and PSNR of 33.5791, as shown in Table V.

VII. DISCUSSION

A. Impact of Aerosol Data

Our method incorporates aerosol data as an additional condition. We conducted relevant ablation experiments to evaluate the impact of aerosol parameters on the model's performance. The experiments utilized Dataset 1-1 and Dataset 1-2 with the aerosol data removed.

The results of the ablation experiments are presented in Table VI, demonstrating a significant improvement when aerosol parameters are included. The SSIM increases by approximately 0.01, and the PSNR improves by 3 dB. Furthermore, additional tests were conducted on atmospheric correction for MSIs. The SSIM increases by approximately 0.1, and the PSNR increases by around 9 dB when using aerosol parameters as a condition. These results

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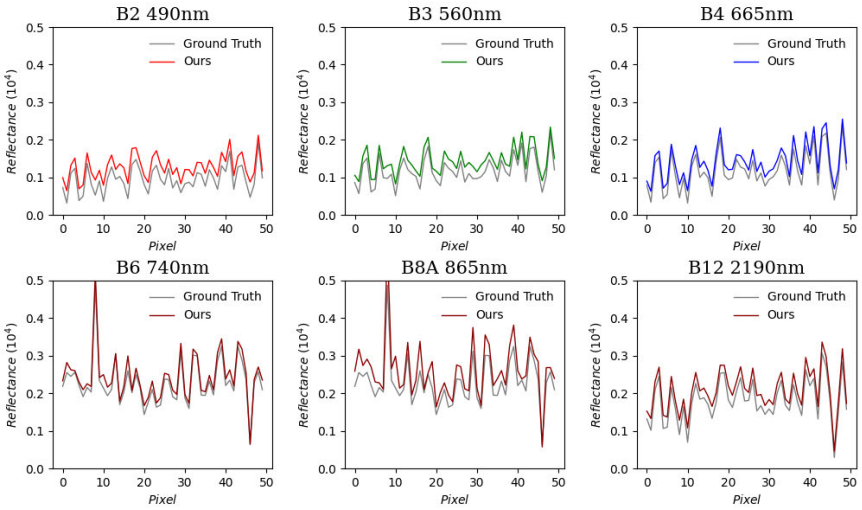


Fig. 15. Example of atmospheric correction reflectance results for six bands in urban area.

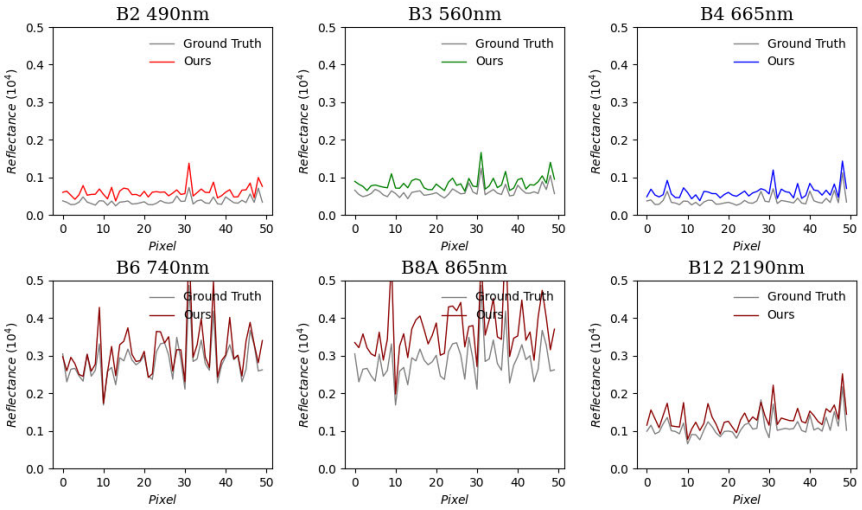


Fig. 16. Example of atmospheric correction reflectance results for six bands in vegetation area.

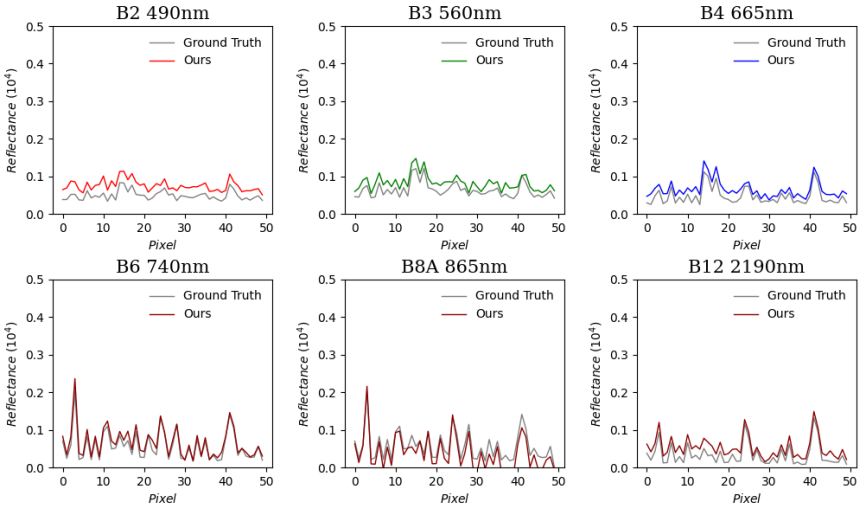


Fig. 17. Example of atmospheric correction reflectance results for six bands in water area.

demonstrate that incorporating aerosol parameters as an additional condition has a positive impact on atmospheric correction for our method.

Further comparison of various quantitative metrics reveals, as depicted in Fig. 20, that the introduction of aerosol parameters as auxiliary conditions improves the atmospheric

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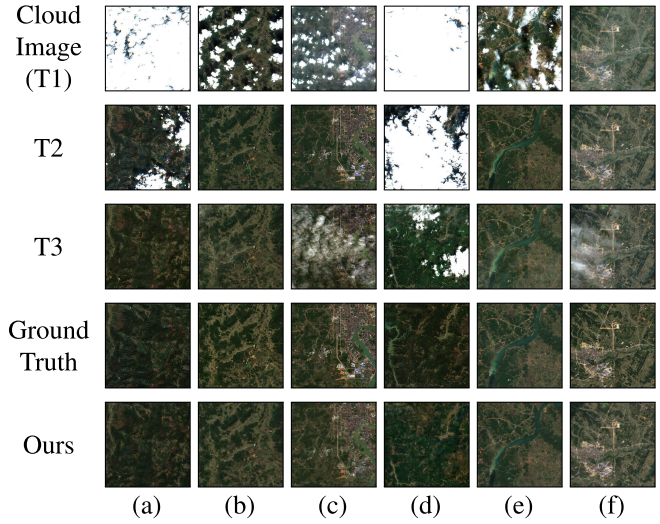


Fig. 18. Cloud removal result.

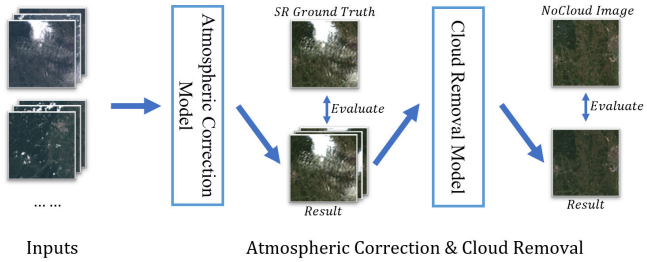


Fig. 19. Two-step process flowchart.

correction performance of both panchromatic and multispectral ground reflectance images. In the absence of aerosol conditions, the performance of multispectral ground reflectance declines compared with panchromatic images due to the need to correct for 12 channels. However, with the addition of aerosol parameter conditions, model performance improves, and all quantitative metrics surpass the correction results of panchromatic images.

The results of further comparison of various quantitative indicators are shown in Fig. 20. It demonstrates that the atmospheric correction performance of both true color and multispectral is improved with the addition of aerosol parameters as auxiliary conditions. The performance of multispectral correction is inferior to that of panchromatic correction without incorporating aerosol conditions. However, the improvement in multispectral atmospheric correction is the most significant with the inclusion of aerosol parameters.

B. Impact of Multitemporal Data on Cloud Removal

To further assess the impact of multitemporal image data on the quality of cloud removal, we conducted separate tests on cloud removal results for remote sensing images with time lengths ranging from 2 to 4, as shown in Table VII. With an increase in the number of input images, all three methods demonstrated varying degrees of performance improvement on the test set of Dataset 2-1, confirming the positive effect of multitemporal data on the remote sensing cloud removal. In addition, our method effectively integrates multiple images to achieve results close to the ground truth. However, when

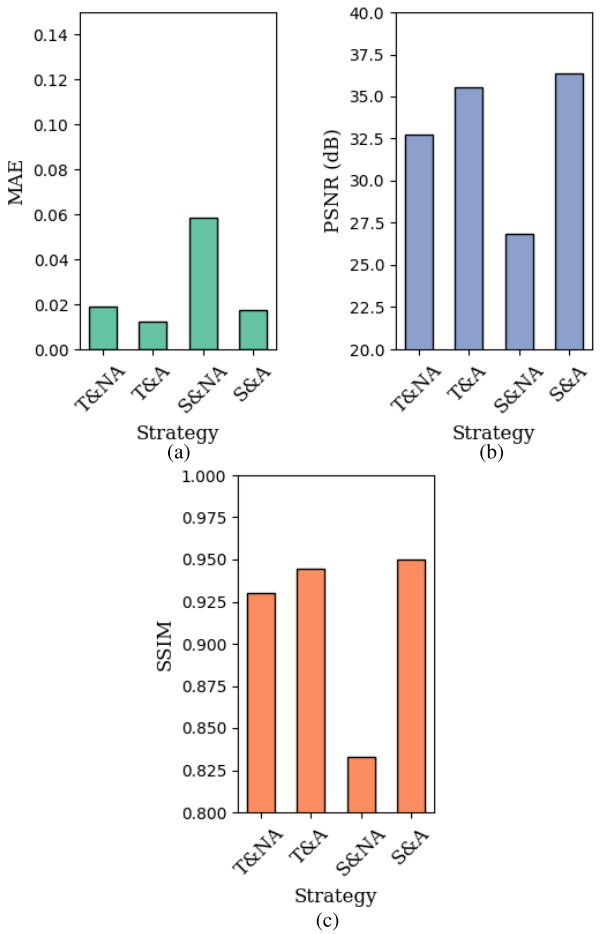


Fig. 20. Comparison of quantitative evaluation results. T represents true color, S represents multispectral, NA represents without aerosol, and A represents with aerosol. (a) MAE. (b) PSNR. (c) SSIM.

setting the sequence length to $L = 5$, the training dataset size is significantly reduced, and the time span between some adjacent image sequences exceeds 30 days, which compromises the consistency of land features. Therefore, sequences with $L > 4$ were not considered.

The charts in Fig. 21(a)–(c), respectively, compare the MAE, PSNR, and SSIM metrics of different models under various time sequence lengths ($L = 2$, $L = 3$, and $L = 4$). They provide a visual representation of the differences in cloud removal performance. Overall, the cloud removal performance improves with an increase in the input sequence length. Considering all three metrics, our method performs slightly better than U-Net at a single time point ($L = 2$). However, its performance progressively improves and surpasses that of the U-Net at $L = 3$ and $L = 4$. U-Net and GAN do not effectively utilize multitemporal data, showing no significant improvement or even a decrease in performance at $L = 3$ and $L = 4$. Quantitative Evaluation Results for different length are shown in Table VII.

C. Comparison

In this section, we selected several atmospheric corrections, image dehazing, and cloud removal methods as comparative experiments. We evaluated the performance of different methods compared with the method proposed in this study for

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TABLE VII
QUANTITATIVE EVALUATION RESULTS FOR s IN MULTITEMPORAL CLOUD REMOVAL

L	Quantitative Metrics								
	SEQ-UNET			SEQ-CGAN			Ours		
	MAE	SSIM	PSNR	MAE	SSIM	PSNR	MAE	SSIM	PSNR
2	0.0296	0.8184	27.9678	0.0451	0.6951	24.599	0.0337	0.9031	29.4419
3	0.0281	0.8308	28.1201	0.0518	0.7707	24.1654	0.0263	0.9728	32.2056
4	0.0294	0.869	28.3151	0.0294	0.8989	27.8352	0.0224	0.9556	36.2122

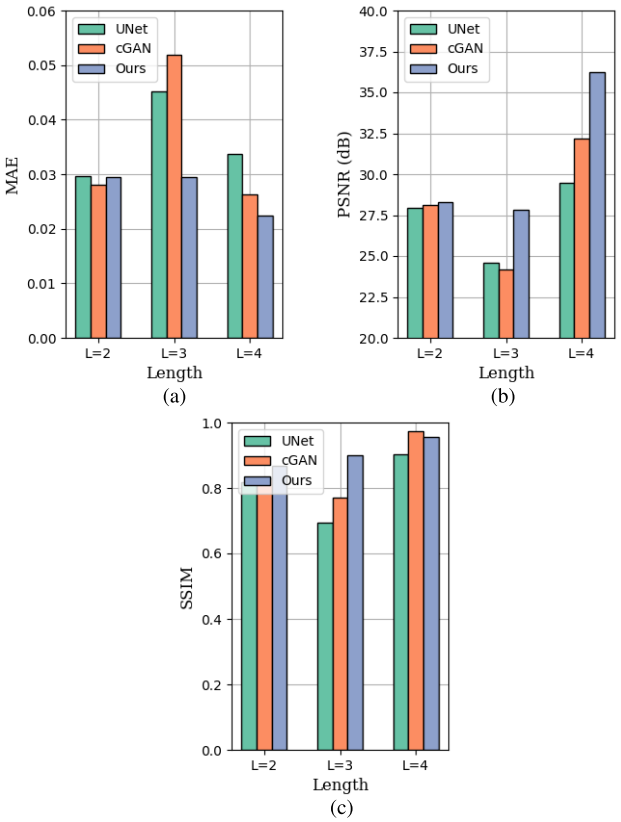


Fig. 21. Comparison of quantitative evaluation results for cloud removal at different lengths of multitemporal data. (a) MAE. (b) PSNR. (c) SSIM.

cloud and haze removal tasks. All comparative experiments used the dataset created in this study to ensure the comparability of the results. To ensure comparability of results, nondeep learning algorithms were tested and evaluated on the test set. Deep learning methods were trained and tested using training and testing sets that were fully consistent with this experiment. The following comparison algorithms were selected for this study.

DCP [29], [30]: The core concept of the DCP for fog removal is that pixels in at least one color channel have very low values, and it uses these dark channel pixels to estimate the fog density in the image, thereby restoring a clear image. Fast line-of-sight atmospheric analysis of spectral hypercubes (FLAASH) Model: Based on the MODTRAN4+ radiative transfer model, the FLAASH atmospheric correction model supports atmospheric correction of MSIs. This study uses the ENVI FLAASH atmospheric correction module. Principled synthetic-to-real dehazing (PSD) [38]: a supervised pretrained fog removal model that considers the physical priors of image defogging. It is fine-tuned unsupervised based on a loss model guided by light scattering and transmission models. Conditional generative adversarial Nets (CGAN) [23], [31], [32], [33], [34], [35]: GANs are generative networks widely used in image generation, image translation, and other tasks. Remote sensing images, often obscured by clouds or fog, can benefit from the prior knowledge prelearned by generative networks. CGAN [36] is an improvement on the original GAN, using the additional information to control image generation. This comparative experiment chose the pix2pix-CGAN network to test the performance of GAN-based generative models in remote sensing image defogging, especially cloud removal. U-Net [20], [21]: Widely used in image semantic segmentation and pattern recognition, U-Net connects the feature maps between the encoder and decoder via skip connections, fully utilizing the image's contextual information for image-to-image tasks. This study uses a U-Net model with an added attention mechanism to test the performance of nongenerative image-to-image models in the remote sensing image defogging and cloud removal. All in one dehazing network (AOD-Net) [30]: based on a CNN network and a reconstructed atmospheric scattering model, this model does not require the complex calculations of traditional atmospheric scattering models and can directly output clear defogged images through the CNN network. In addition, the multitemporal cloud removal method (Mosaic) [37] was added for cloud removal tasks. This method is based on remote sensing images obtained at different times from the same area, using the dynamic characteristics of cloud movement to replace and optimize cloud coverage in multiple remote sensing images to achieve the goal of image cloud removal.

1) Atmospheric Correction Comparison: We quantitatively evaluated the models using the test sets from the corresponding task datasets with three evaluation metrics: MAE, SSIM, and PSNR. Table VIII summarizes the overall metric results for the TCI and multispectral test sets.

From the results in Table VIII, it can be seen that the proposed model in this study achieves an average SSIM of 0.94 and a PSNR of 35.5 dB when compared with ground truth in the task of full-color image (TCI) correction. Our method shows an improvement of approximately 50%, 9%, and 29% in terms of MAE, SSIM, and PSNR compared with other deep learning algorithms, such as U-Net and AOD-Net, respectively. Compared with traditional nondeep learning algorithms such as DCP, our method demonstrates an improvement of approximately 17 times, 25%, and 90% in terms of MAE, SSIM, and PSNR, respectively.

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TABLE VIII
QUANTITATIVE EVALUATION RESULTS OF COMPARATIVE METHODS

Method	Quantitative Metrics					
	TCI Correction			MSI Correction		
	MAE	SSIM	PSNR	MAE	SSIM	PSNR
CGAN	0.0331	0.7959	27.3727	0.0337	0.8355	26.5198
UNet	0.0241	0.8902	30.031	0.0226	0.8775	31.2302
PSD	0.0314	0.8148	26.5071	-	-	-
AOD-Net	0.0495	0.7494	22.8891	0.0486	0.6235	24.1735
FLAASH	0.2082	0.5079	15.3267	0.1508	0.7614	16.5725
DCP	0.2391	0.8231	20.2063	-	-	-
Ours	**0.0125**	**0.9445**	**35.5221**	**0.0174**	**0.9499**	**36.3547**

[1] Bold indicates the best results, underline indicates the second best results.

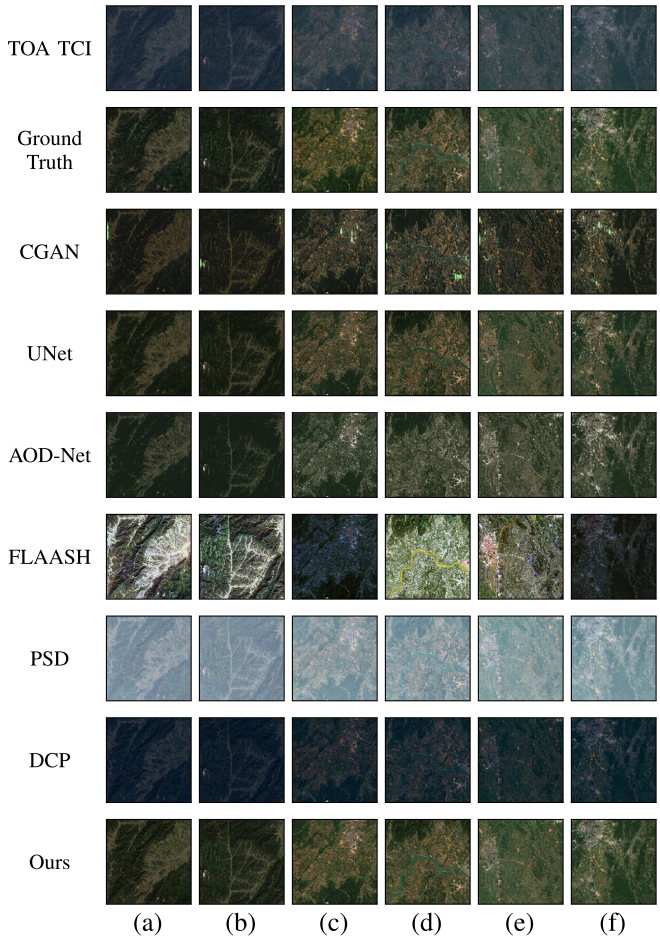


Fig. 22. Sample results of comparative true color atmosphere correction method.

Our method achieves an average SSIM of 0.94 and a PSNR of 36.3 dB in the task of multispectral correction. There is an improvement of approximately 10% and 24% compared with other deep learning algorithms. Compared with traditional algorithms, there is an improvement of approximately 17% and 68% in terms of SSIM and PSNR, respectively.

Three types of TCIs are presented as samples in Fig. 22. With mountainous and forested areas as the main regions, urban and agricultural lands as the main areas, and including rivers and reservoirs. It can be seen traditional algorithms such as DCP and FLAASH exhibit significant deviations. It causes problems such as overbrightness, darkness, or color bias. PSD performs poorly in satellite remote sensing image processing.

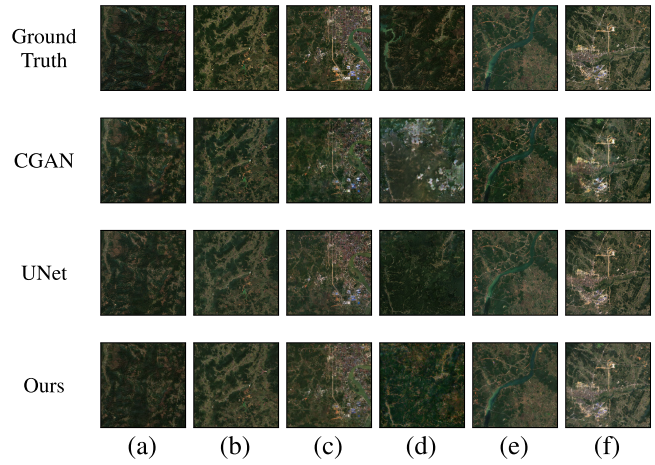


Fig. 23. Sample results of comparative cloud removal method.

U-Net, CGAN, AOD-Net, and our method all accurately reflect the true spectral information. However, AOD-Net exhibits a more significant loss of ground texture information compared with other models. The results of the CGAN model show partial distortion, mainly concentrated in areas with small-scale cloud cover and complex urban features. Due to the specific nature of the DCP algorithm and PSD model, we only tested their correction performance on TCIs.

Quantitative evaluation results of atmospheric correction for three types (vegetation, urban, and water area) of regions are presented in Table IX. For vegetation-covered areas, our proposed method achieves the best performance in the SSIM, indicating its superior ability to preserve image texture compared to other contrast models. Although slightly inferior in PSNR compared with some deep learning methods such as CGAN, the achieved PSNR of 26.7 still demonstrates its excellent performance.

The quantitative evaluation of atmospheric correction for urban areas is shown in the urban section of Table IX. Our method achieves an SSIM of 0.98 and a PSNR of 33.9, higher than other comparative methods, indicating good performance in atmospheric correction for urban areas. Compared with other methods, our approach exhibits better performance in restoring ground textures, especially in areas, such as buildings and road networks. In addition, the atmospheric correction results are closest to the ground-truth SR image, with an absolute average error of approximately 1.8.

The quantitative evaluation results for water areas are shown in the water section of Table IX. Our method achieves an overall absolute average error of 1.5%, with an SSIM of 0.96 and a PSNR of 33.9 dB. Compared with the comparative methods, our approach also achieves the best performance.

Our study further compares the latest relevant models with the proposed model on the dataset introduced in this research for atmospheric correction. The following are the latest comparison models selected in this article: DiffCR [38], DDPM-Enhancement [39], and guided diffusion (palette) [40], which employ diffusion models; CR-TS [41] and STGAN [23]; UnCRtainTS [42], which utilizes a U-Net-like encoder–decoder structure; and GLF-CR [43], a convolutional

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TABLE IX
QUANTITATIVE EVALUATION RESULTS OF ATMOSPHERE CORRECTION IN DIFFERENT REGIONS

		(a)GT	(c)UNet	(d)cGan	(e)DCP	(f)FLAASH	(g)PSD	(h)AOD-Net	(b)Ours
Vegetation	MAE	0	0.0296	0.0278	0.0543	0.1551	0.4296	**0.0275**	0.0341
	SSIM	1	0.6628	0.663	0.6174	0.2116	0.2855	0.5886	**0.7162**
	PSNR	Inf	27.3579	**27.8675**	23.7118	14.382	7.2144	0.5886	26.7128
	R	1	**0.6011**	0.5568	0.2007	0.2559	0.2184	0.5268	0.5265
Urban	MAE	0	0.0742	0.1413	0.1797	0.1877	0.2661	0.0649	**0.0187**
	SSIM	1	0.9254	0.6691	0.6478	0.5766	0.8597	0.6009	**0.9806**
	PSNR	Inf	21.4847	14.5708	13.8578	13.4734	11.0159	21.1059	**33.9455**
	R	1	0.9661	0.598	0.8195	0.7644	0.8181	0.8292	**0.9916**
Water	MAE	0	0.0331	0.0491	0.1123	0.2637	0.3188	0.05445	**0.015**
	SSIM	1	0.8998	0.7924	0.7282	0.4356	0.6792	0.6378	**0.9667**
	PSNR	Inf	27.0986	24.0386	17.4569	10.2062	9.6488	22.8314	**33.9455**
	R	1	0.9592	0.9148	0.7592	0.6871	0.7361	0.8042	**0.9867**

[1] Bold indicates the best results, underline indicates the second best results.

TABLE X
RESULTS FOR RADIATIVE TRANSFER ATMOSPHERIC CORRECTION

Method	MAE	RMSE	SSIM	PSNR	R	SAM	FID	LPIPS	NIQE
DiffCR(2024)	0.0592	0.0703	0.77	22.9614	0.9306	0.1476	151.7841	0.0918	16.4781
DDPM-Enhancement(2024)	0.103	0.118	0.66	18.2944	0.9556	0.3315	239.9036	0.1242	22.5419
UnCRtainTS(2023)	0.0219	0.0324	0.89	30.4474	0.9722	0.0535	77.6401	0.0485	12.1641
U-TILISE(2023)	0.0173	0.0299	0.88	30.1724	**0.9836**	**0.0484**	134.0371	0.1331	18.5689
Guided Diffusion(Palette)(2022)	0.5387	0.5535	0.26	5.1398	0.2225	0.4336	207.5495	0.2117	20.9431
CR-TS Net(2022)	0.0451	0.0712	0.85	28.8216	0.8677	0.0811	151.3128	0.1412	15.0667
STGAN(2020)	0.0781	0.1046	0.61	24.1565	0.6072	0.1931	140.9231	0.1197	12.5264
Ours	**0.0172**	**0.0204**	**0.93**	**31.5535**	0.9772	0.0897	**72.3525**	**0.0249**	**8.8285**

[1] Bold indicates the best results, underline indicates the second best results.

TABLE XI
QUANTITATIVE COMPARISON OF CLOUD REMOVAL METHODS

Method	Quantitative Metrics		
	MAE	SSIM	PSNR
SEQ-CGAN	0.0294	0.8989	27.8352
SEQ-UNet	0.0294	0.869	28.3151
MULTI-TEMPORAL MOSAIC	**0.1995**	0.9114	29.2252
Ours	0.0224	**0.9556**	**36.2122**

[1] Bold indicates the best results, underline indicates the second best results.

neural network architecture. The testing conditions are identical to the experimental methods used in this study. The results are shown in Table X.

The test results after incorporating the Fréchet inception distance (FID), learned perceptual image patch similarity (LPIPS), and naturalness image quality evaluator (NIQE) evaluation metrics are shown in Table X. The model proposed in this study achieves optimal results in FID, LPIPS, NIQE, and MAE. The correlation coefficient R is nearly consistent with that of the SOTA methods; the SAM metric is of the same order of magnitude as the best SOTA results; and for the other seven metrics, the results from this study are significantly better.

D. Cloud Removal

Fig. 23 illustrates the cloud removal results using a time length of 4 for the image sequence. Due to the stable revisit period of satellite remote sensing images for the same area, some image sequences contain at least one image with almost no cloud cover (coverage ratio less than 2%). The model can effectively infer cloud-covered areas and replace them. It can achieve optimal results for such multitemporal data. When all image sequences are covered by thick or thin clouds, our model can still effectively infer ground features in completely cloud-covered areas, as shown in Fig. 23(d) and (f). The quantitative evaluation test results of our method and relevant comparative methods on the test set of dataset 2-1 are shown in Table XI. Our method achieves approximately 0.95 SSIM and 36-dB PSNR in the multitemporal cloud removal task.

This study further compares the latest relevant models with the proposed model on the dataset introduced in this research for multitemporal cloud removal. The comparison models are as follows: DiffCR [38], DDPM-Enhancement [39], and guided diffusion (palette) [40], which employ diffusion models; CR-TS [41] and STGAN [23]; UnCRtainTS [42], which utilizes a U-Net-like encoder–decoder structure; and GLF-CR [43], a convolutional neural network architecture. The testing conditions are identical to the experimental methods used in this study. The results are shown in Table XII.

The test results after incorporating the FID, LPIPS, and NIQE evaluation metrics are shown in Table XII. The model proposed in this study achieves optimal results in metrics, such as FID, LPIPS, and NIQE. The MAE evaluation metric of the model is nearly identical to that of the best method. Overall, the proposed model achieves the best results in atmospheric correction tasks compared with the latest SOTA methods.

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TABLE XII
RESULTS FOR RADIATIVE TRANSFER MULTITEMPORAL CLOUD REMOVAL

Method	MAE	PSNR	SSIM	FID	LPIPS	NIQE
DiffCR(2024)	0.0277	29.1484	0.89	115.4649	0.0641	17.5371
U-TILISE(2023)	0.0248	29.1206	0.9	94.3238	0.0393	16.0904
UnCRtainTS(2023)	**0.0131**	28.3164	0.93	79.6591	0.0795	18.1442
Guided Diffusion(Palette)(2022)	0.5369	5.1579	0.27	275.4116	0.6806	23.1858
CR-TS Net(2022)	0.0451	28.8216	0.85	60.7433	0.084	14.6085
STGAN(2020)	0.0465	28.6217	0.84	63.0472	0.0939	15.0667
Ours	0.0154	**32.2056**	**0.94**	**58.9701**	**0.0361**	**12.7834**

[1] Bold indicates the best results, underline indicates the second best results.

TABLE XIII
RESULT ON RICE-DATASET

Method	RICE-1		RICE-2	
	PSNR	SSIM	PSNR	SSIM
Basline (GAN)	31.03	0.91	30.04	0.8
SpA GAN (2020)	30.23	0.95	28.37	**0.91**
DiffBIR (2023)	28.26	0.71	28.06	0.81
RSHazeDiff (2024)	29.65	0.91	27.91	0.9
IDF-CR (2024)	31.19	0.95	30.89	0.9
Ours	**32.38**	**0.96**	**32.29**	0.9

[1] Bold indicates the best results, underline indicates the second best results.

TABLE XIV
RESULT ON SEN12MS-CR-TS DATASET

Method	RMSE	SAM	PSNR	SSIM
least cloudy	0.079	12.204	—	0.815
STGAN (2020)	0.057	12.548	25.42	0.818
CR-TS Net (2022)	0.051	10.657	26.68	0.836
UnCRtainTS (2023)	0.051	10.16	27.84	**0.866**
SeqDMs (2023)	**0.045**	12.777	28.07	0.827
Ours	0.052	**10.027**	**29.32**	0.855

[1] Bold indicates the best results, underline indicates the second best results.

E. Test Results on Open-Source Datasets

Considering the broad applicability of the study area and further validating the model's generalization ability, this article tests the proposed method using the widely adopted open-source datasets RICE-Dataset and SEN12MS-CR. Comparisons were made with the respective SOTA methods, and the results demonstrate that the proposed method achieves the best performance metrics. This article uses two types of data from RICE-Dataset [44], including thin clouds (RICE-1) and thick clouds (RICE-2). Among the comparison methods, IDF-CR [45], RSHazeDiff [46], and DiffBIR [47] utilize diffusion models, while Baseline [44] and SpA GAN [35] use GANs. The results are shown in Table XIII.

It can be seen in the result that our proposed method shows a clear advantage on the RICE-1 dataset. The PSNR metric demonstrates a significant advantage, while the SSIM results are comparable to the current public results on the RICE-2 dataset. Overall, the proposed method's performance metrics are significantly better than others.

SEN12MS-CR-TS [41] dataset was used to test the cloud removal performance on multitemporal multispectral remote sensing data. Among the comparison methods, least cloudy is a replacement method, CR-TS [41] is the baseline provided by the dataset, SeqDMs [48] uses a diffusion model, STGAN [23] uses a GAN, and UnCRtainTS [42] utilizes a U-Net-like encoder–decoder structure. The results are shown in Table XIV.

Our proposed method shows a significant advantage in SAM and PSNR metrics, with SSIM results nearly identical to the SOTA methods on SEN12MS-CR-TS dataset. Overall, our proposed method achieves the best results.

TABLE XV
MODEL PERFORMANCE STATISTICS

Model	Parameters	GFLOPs
Guided Diffusion	62M	410
SR3	61M	361
Pix2pix-CGAN	57M	56
UNet	110M	540
PSD	33M	90
Ours	80M	487

TABLE XVI
DIFFUSION STEPS

T	Atmospheric Correction			Cloud Removal(Length=3)			Inference
	MAE	PSNR	SSIM	MAE	PSNR	SSIM	
10	0.027	0.8834	29.19	0.121	0.687	19.28	3s
25	0.021	0.8706	31.17	0.049	0.861	26.42	9s
50	**0.014**	0.9379	**34.61**	0.025	0.925	**31.39**	15s
100	0.015	**0.9335**	34.02	**0.021**	**0.908**	32.29	34s

Bold indicates the best results.

F. Performance

This section provides statistics on the parameters and floating point operations (FLOPs) of several comparison methods used in experiments and some diffusion models widely used as baseline models to reflect model performance. Our model needs to significantly outperform other comparison methods even though its parameters are slightly larger than those of other models. Generally, better performance usually means a larger model. Given that our task does not have high real-time requirements, a larger model size is acceptable for this study. Statistics for GFLOPs are shown in Table XV.

G. Diffusion Steps

Our method was optimized using DDIM. Considering the computational cost of the reverse process, we generated images at a resolution of 256 × 256 with eight images per batch. The results for 10, 25, 50, and 100 steps are shown in the table.

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The test results in Table XVI indicate a positive correlation between performance and the number of steps. However, the improvement at 100 steps compared with 50 steps is not significant. Considering inference performance, this study opts to use $T = 50$ for testing.

VIII. Conclusion

Atmospheric aerosols and clouds are the main factors affecting the acquisition of high-quality satellite remote sensing ground imagery during the radiometric calibration process. This study proposed a method based on a conditional guided diffusion model. It achieves unified atmospheric correction and cloud removal through a redesigned model for reverse denoising and conditional guidance network structure. This method effectively integrates key information on the quality factors of remote sensing data, such as multispectral and aerosol imagery.

The proposed method was evaluated using Sentinel-2 multispectral remote sensing imagery. For atmospheric correction tasks on TCIs, our method achieved MAE, SSIM, and PSNR metrics of 0.0125, 0.9445, and 35.5221 dB. For the atmospheric correction task in MSIs, the method achieved MAE, SSIM, and PSNR metrics of 0.0174, 0.9499, and 36.3547 dB. For multitemporal true color cloud removal tasks, the method achieved MAE, SSIM, and PSNR metrics of 0.0224, 0.9556, and 36.2122 dB. Compared with other models and methods, the proposed method achieved the best evaluation results.

However, the proposed method still has some shortcomings and requires further optimization. To further enhance the model's performance, future research plans to use a multitask learning approach to perform atmospheric correction and cloud removal tasks. In addition, since this study used Sentinel-2 as the data source, the next step is to validate the model's generalizability and scalability by applying it to different satellite remote sensing imagery datasets.

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